Core concurrent circular buffer

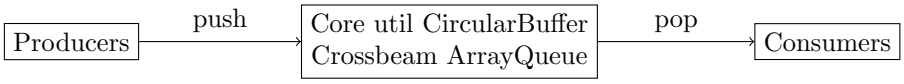
Created-at: 2026-09-13 03:42 +02:00. ETA: 25 minutes for integration, targeted correctness checks and publication; no performance run.

Scope and authorization. The user explicitly requests a qualified multithreaded circular buffer in Core, reusing Rust facilities where available. Expose Crossbeam's existing bounded MPMC implementation through Core util; do not maintain a second atomic algorithm. This request authorizes the scoped protected source and dependency edits. No existing callers require migration.

Files. hymeko_core/Cargo.toml, Cargo.lock, hymeko_core/src/util/mod.rs, hymeko_core/tests/circular_buffer.rs, SYMBOLS.md if generated output changes, this four-format plan, and the ignored reports/2026-09-13-concurrent-circular-buffer.md. Protected items: hymeko_core source and manifest.

Contract. Re-export ArrayQueue as CircularBuffer, with exactly pinned crossbeam-queue 0.3.14. Positive fixed capacity; push returns the rejected value when full; force_push explicitly evicts the oldest value; pop returns None when empty. Share across threads for Send payloads. Observations of length do not reserve slots. No fairness, real-time deadline, crash recovery or formal universal correctness claim. Allocation can fail; zero/excessive capacity can panic.

Qualification. Dedicated integration target only: FIFO and repeated slot reuse against VecDeque at capacities 1, 2, 3 and 7; overflow and eviction; exact ownership/drop accounting; concurrent delivery and producer order; small recorded operation histories checked against a sequential FIFO model with an iterative linearizability search and positive/negative oracle controls. No profiling, throughput loops, latency assertions or workspace aggregate. Run scoped Clippy, formatting, symbols, preflight and duplication checks.

Resources and risks. No performance budget is measured or claimed. Correctness fixtures are small, bounded workloads. Production risks include contention, starvation and unhandled rejection; callers must explicitly choose retry/backpressure or eviction. Tests do not exhaust hardware memory models, all schedules, stamp rollover or every target architecture.

Rollback. Revert this task's commit; no serialized format or existing behavior changes. Preserve pre-existing untracked LaTeX helper files.